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Pinned group summary: SL-n3

Pinning-Sympy automated output

July 13, 2026

Contents

1 Basic group attributes

Group	SL-n3
Matrix size	3×3
Rank	2
Defining equations	$\det(X) = 1$

1.1 Lie algebra

The defining equations for the Lie algebra are: $\mathrm{tr}(X) = 0$
A generic element of the Lie algebra is:

$$\begin{bmatrix} x_{00} & x_{01} & x_{02} \\ x_{10} & x_{11} & x_{12} \\ x_{20} & x_{21} & -x_{00} - x_{11} \end{bmatrix}$$

1.2 Torus

The torus has rank 2. A generic element of the torus is

$$\begin{bmatrix} t_0 & 0 & 0 \\ 0 & t_1 & 0 \\ 0 & 0 & \frac{1}{t_0 t_1} \end{bmatrix}$$

The rows of the matrix below are trivial characters:

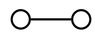
$$\begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$$

2 Root system

Dynkin type	A_2
Irreducible	True
Reduced	True
Simply laced	True
Number of roots	6

2.1 Dynkin diagram

The root system is irreducible and has Dynkin diagram:



2.2 Cartan matrix

The root system is irreducible and has Cartan matrix:

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

2.3 Roots

Definition 2.1. A root $\alpha \in \Phi$ is **multipliable** if $2\alpha \in \Phi$.

Root	**Simple**	**Sign**	**Multipliable**
$(-1,\ 0,\ 1)$	No	$-$	No
$(-1,\ 1,\ 0)$	No	$-$	No
$(0,\ -1,\ 1)$	No	$-$	No
$(0,\ 1,\ -1)$	Yes	$+$	No
$(1,\ -1,\ 0)$	Yes	$+$	No
$(1,\ 0,\ -1)$	No	$+$	No

2.4 Coroots

Root	**Coroot**
$(-1,\ 0,\ 1)$	$(-1,\ 0,\ 1)$
$(-1,\ 1,\ 0)$	$(-1,\ 1,\ 0)$
$(0,\ -1,\ 1)$	$(0,\ -1,\ 1)$
$(0,\ 1,\ -1)$	$(0,\ 1,\ -1)$
$(1,\ -1,\ 0)$	$(1,\ -1,\ 0)$
$(1,\ 0,\ -1)$	$(1,\ 0,\ -1)$

2.5 Linear combinations of roots

Definition 2.2. For two roots α, β in a root system Φ, the set of positive integral linear combinations of α, β that are also roots is denoted $(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$.

α	β	Pairs (i, j)
$(-1,\ 0,\ 1)$	$(0,\ 1,\ -1)$	$(1, 1)$
$(-1,\ 0,\ 1)$	$(1,\ -1,\ 0)$	$(1, 1)$
$(-1,\ 1,\ 0)$	$(0,\ -1,\ 1)$	$(1, 1)$
$(-1,\ 1,\ 0)$	$(1,\ 0,\ -1)$	$(1, 1)$
$(0,\ -1,\ 1)$	$(1,\ 0,\ -1)$	$(1, 1)$
$(0,\ 1,\ -1)$	$(1,\ -1,\ 0)$	$(1, 1)$

3 Root spaces

3.1 Dimensions

The table below lists each root along with the dimension of the associated root space (and root subgroup).

Root (α)	**Dimension of root space** (d_α)
$(-1,\ 0,\ 1)$	1
$(-1,\ 1,\ 0)$	1
$(0,\ -1,\ 1)$	1
$(0,\ 1,\ -1)$	1
$(1,\ -1,\ 0)$	1
$(1,\ 0,\ -1)$	1

3.2 Root space table

The table below lists each root along with a generic element of the associated root space (inside the Lie algebra). Include a definition here? INCOMPLETE

Root	**Generic element of root space**
$(-1,\ 0,\ 1)$	$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ u_0 & 0 & 0 \end{bmatrix}$
$(-1,\ 1,\ 0)$	$\begin{bmatrix} 0 & 0 & 0 \\ u_0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$
$(0,\ -1,\ 1)$	$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & u_0 & 0 \end{bmatrix}$
$(0,\ 1,\ -1)$	$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & u_0 \\ 0 & 0 & 0 \end{bmatrix}$
$(1,\ -1,\ 0)$	$\begin{bmatrix} 0 & u_0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$
$(1,\ 0,\ -1)$	$\begin{bmatrix} 0 & 0 & u_0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

4 Root subgroups

4.1 Root subgroup table

The table below lists each root along with a generic element of the associated root subgroup.

Root	Generic element of root subgroup
$(-1,\ 0,\ 1)$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ u_0 & 0 & 1 \end{bmatrix}$
$(-1,\ 1,\ 0)$	$\begin{bmatrix} 1 & 0 & 0 \\ u_0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
$(0,\ -1,\ 1)$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & u_0 & 1 \end{bmatrix}$
$(0,\ 1,\ -1)$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & u_0 \\ 0 & 0 & 1 \end{bmatrix}$
$(1,\ -1,\ 0)$	$\begin{bmatrix} 1 & u_0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
$(1,\ 0,\ -1)$	$\begin{bmatrix} 1 & 0 & u_0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

4.2 Homomorphism property

For each root $\alpha \in \Phi$, the associated root subgroup map $X_\alpha : V_\alpha \to G$ is a pseudo-homomorphism with respect to the additive structure on V_α and the group operation (i.e. matrix multiplication) in G. That is, for any $u, v \in V_\alpha$,

$$X_\alpha(u) \cdot X_\alpha(v) = X_\alpha(u+v) \cdot \prod_{\substack{i>1 \\ i\alpha\in\Phi}} X_{i\alpha}\Big(q_\alpha^i(u,v)\Big) \tag{1}$$

for some homogeneous polynomial maps $q_\alpha^i : V_\alpha \times V_\alpha \to V_{i\alpha}$. Thus X_α is a group homomorphism if and only the product on the right hand side reduces to 1. The quantities $q_\alpha^i(u,v)$ are called **homomorphism defect coefficients**. For any root α of a root system Φ of any group considered in the Pinning-Sympy package, the set

$$\{i\alpha \in \Phi : i \in \mathbb{Z}_{\geq 2}\}$$

is either empty or a singleton set consisting of $\{2\alpha\}$. The non-empty case only arises when Φ is a non-reduced root system (of Dynkin type BC), and when α is a multipliable root.

Since the root system for this group is reduced, there are no multipliable roots and the product is always empty, hence each X_α is a group homomorphism there are no homomorphism defect coefficients.

5 Commutators

The Steinberg/Chevalley commutator formula says that for two roots $\alpha, \beta \in \Phi$, the commutator of two root group elements $X_\alpha(u)$ and $X_\beta(v)$ is controlled by the structure of Φ, in particular by which positive integer linear combinations $i\alpha + j\beta$ are elements of Φ. Recall that

$$(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$$

Specifically, we have the commutator formula

$$\Big[X_\alpha(u), X_\beta(v)\Big] = \prod_{(\alpha,\beta)} X_{i\alpha+j\beta}\Big(N_{ij}^{\alpha\beta}(u,v)\Big)$$

for some homogeneous polynomial functions $N_{ij}^{\alpha\beta} : V_\alpha \times V_\beta \to V_{i\alpha+j\beta}$. We repeat here the table of linear combinations of roots. The quantity $N_{ij}^{\alpha\beta}(u,v)$ is called a **commutator coefficient**. The table below gives these coefficients.

α	β	i	j	$i\alpha + j\beta$	$N_{ij}^{\alpha\beta}(u,v)$
$(-1,\ 0,\ 1)$	$(0,\ 1,\ -1)$	1	1	$(-1,\ 1,\ 0)$	$-u_0v_0$
$(-1,\ 0,\ 1)$	$(1,\ -1,\ 0)$	1	1	$(0,\ -1,\ 1)$	u_0v_0
$(-1,\ 1,\ 0)$	$(0,\ -1,\ 1)$	1	1	$(-1,\ 0,\ 1)$	$-u_0v_0$
$(-1,\ 1,\ 0)$	$(1,\ 0,\ -1)$	1	1	$(0,\ 1,\ -1)$	u_0v_0
$(0,\ -1,\ 1)$	$(-1,\ 1,\ 0)$	1	1	$(-1,\ 0,\ 1)$	u_0v_0
$(0,\ -1,\ 1)$	$(1,\ 0,\ -1)$	1	1	$(1,\ -1,\ 0)$	$-u_0v_0$
$(0,\ 1,\ -1)$	$(-1,\ 0,\ 1)$	1	1	$(-1,\ 1,\ 0)$	u_0v_0
$(0,\ 1,\ -1)$	$(1,\ -1,\ 0)$	1	1	$(1,\ 0,\ -1)$	$-u_0v_0$
$(1,\ -1,\ 0)$	$(-1,\ 0,\ 1)$	1	1	$(0,\ -1,\ 1)$	$-u_0v_0$
$(1,\ -1,\ 0)$	$(0,\ 1,\ -1)$	1	1	$(1,\ 0,\ -1)$	u_0v_0
$(1,\ 0,\ -1)$	$(-1,\ 1,\ 0)$	1	1	$(0,\ 1,\ -1)$	$-u_0v_0$
$(1,\ 0,\ -1)$	$(0,\ -1,\ 1)$	1	1	$(1,\ -1,\ 0)$	u_0v_0

5.1 Commutator identities

Below is an explicit formulation of the commutator formula identities.

CommutatorIdentitiesPlaceholder

6 Weyl group

Background theoretical info on Weyl group, especially as realized inside the group, including info on conjugation between root subgroups INCOMPLETE

6.1 Weyl group elements

The table below lists roots along with an associated Weyl group element.

Root	**Weyl group element**
$(-1,\ 0,\ 1)$	$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$
$(-1,\ 1,\ 0)$	$\begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
$(0,\ -1,\ 1)$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$
$(0,\ 1,\ -1)$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$
$(1,\ -1,\ 0)$	$\begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
$(1,\ 0,\ -1)$	$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$

6.2 Weyl group conjugation coefficients

The table below lists roots along with the coefficients obtain in performing conjugation by an associated Weyl element.

Root (α)	**Root (β)**	$\gamma = \sigma_\alpha(\beta)$	**Weyl conjugation coefficient**
$(-1,\ 0,\ 1)$	$(-1,\ 0,\ 1)$	$(1,\ 0,\ -1)$	$-u_0$
$(-1,\ 0,\ 1)$	$(-1,\ 1,\ 0)$	$(0,\ 1,\ -1)$	$-u_0$
$(-1,\ 0,\ 1)$	$(0,\ -1,\ 1)$	$(1,\ -1,\ 0)$	u_0
$(-1,\ 0,\ 1)$	$(0,\ 1,\ -1)$	$(-1,\ 1,\ 0)$	u_0
$(-1,\ 0,\ 1)$	$(1,\ -1,\ 0)$	$(0,\ -1,\ 1)$	$-u_0$
$(-1,\ 0,\ 1)$	$(1,\ 0,\ -1)$	$(-1,\ 0,\ 1)$	$-u_0$
$(-1,\ 1,\ 0)$	$(-1,\ 0,\ 1)$	$(0,\ -1,\ 1)$	$-u_0$
$(-1,\ 1,\ 0)$	$(-1,\ 1,\ 0)$	$(1,\ -1,\ 0)$	$-u_0$
$(-1,\ 1,\ 0)$	$(0,\ -1,\ 1)$	$(-1,\ 0,\ 1)$	u_0
$(-1,\ 1,\ 0)$	$(0,\ 1,\ -1)$	$(1,\ 0,\ -1)$	u_0
$(-1,\ 1,\ 0)$	$(1,\ -1,\ 0)$	$(-1,\ 1,\ 0)$	$-u_0$
$(-1,\ 1,\ 0)$	$(1,\ 0,\ -1)$	$(0,\ 1,\ -1)$	$-u_0$
$(0,\ -1,\ 1)$	$(-1,\ 0,\ 1)$	$(-1,\ 1,\ 0)$	$-u_0$
$(0,\ -1,\ 1)$	$(-1,\ 1,\ 0)$	$(-1,\ 0,\ 1)$	u_0
$(0,\ -1,\ 1)$	$(0,\ -1,\ 1)$	$(0,\ 1,\ -1)$	$-u_0$
$(0,\ -1,\ 1)$	$(0,\ 1,\ -1)$	$(0,\ -1,\ 1)$	$-u_0$
$(0,\ -1,\ 1)$	$(1,\ -1,\ 0)$	$(1,\ 0,\ -1)$	u_0
$(0,\ -1,\ 1)$	$(1,\ 0,\ -1)$	$(1,\ -1,\ 0)$	$-u_0$
$(0,\ 1,\ -1)$	$(-1,\ 0,\ 1)$	$(-1,\ 1,\ 0)$	$-u_0$
$(0,\ 1,\ -1)$	$(-1,\ 1,\ 0)$	$(-1,\ 0,\ 1)$	u_0
$(0,\ 1,\ -1)$	$(0,\ -1,\ 1)$	$(0,\ 1,\ -1)$	$-u_0$
$(0,\ 1,\ -1)$	$(0,\ 1,\ -1)$	$(0,\ -1,\ 1)$	$-u_0$
$(0,\ 1,\ -1)$	$(1,\ -1,\ 0)$	$(1,\ 0,\ -1)$	u_0
$(0,\ 1,\ -1)$	$(1,\ 0,\ -1)$	$(1,\ -1,\ 0)$	$-u_0$
$(1,\ -1,\ 0)$	$(-1,\ 0,\ 1)$	$(0,\ -1,\ 1)$	$-u_0$
$(1,\ -1,\ 0)$	$(-1,\ 1,\ 0)$	$(1,\ -1,\ 0)$	$-u_0$
$(1,\ -1,\ 0)$	$(0,\ -1,\ 1)$	$(-1,\ 0,\ 1)$	u_0
$(1,\ -1,\ 0)$	$(0,\ 1,\ -1)$	$(1,\ 0,\ -1)$	u_0
$(1,\ -1,\ 0)$	$(1,\ -1,\ 0)$	$(-1,\ 1,\ 0)$	$-u_0$
$(1,\ -1,\ 0)$	$(1,\ 0,\ -1)$	$(0,\ 1,\ -1)$	$-u_0$
$(1,\ 0,\ -1)$	$(-1,\ 0,\ 1)$	$(1,\ 0,\ -1)$	$-u_0$
$(1,\ 0,\ -1)$	$(-1,\ 1,\ 0)$	$(0,\ 1,\ -1)$	$-u_0$
$(1,\ 0,\ -1)$	$(0,\ -1,\ 1)$	$(1,\ -1,\ 0)$	u_0
$(1,\ 0,\ -1)$	$(0,\ 1,\ -1)$	$(-1,\ 1,\ 0)$	u_0
$(1,\ 0,\ -1)$	$(1,\ -1,\ 0)$	$(0,\ -1,\ 1)$	$-u_0$
$(1,\ 0,\ -1)$	$(1,\ 0,\ -1)$	$(-1,\ 0,\ 1)$	$-u_0$

6.3 Weyl group conjugation identities

Below is an explicit formulation of the Weyl group conjugation identities.

WeylConjugationIdentitiesPlaceholder